

EDUCATION

RTU MIREA — 2nd year

2025 – 2029 · Moscow

Field of study – "Information Systems and Technologies", bachelor's degree (09.03.02). Study program – "Cross-platform Business Application Development".

Additional Education

- "Cybersecurity of Data, OS and Applications" – RTU MIREA elite training (2025 – 2026)
- Online courses from the Sirius Educational Center: "Introduction to C++ Programming", "Introduction to ML", "Deep Learning (DL)", "Natural Language Processing (NLP)"

SKILLS

PROGRAMMING LANGUAGES

C++ 17/20 · Python 3.x · HTML5 · CSS3 · Go · SQL · JavaScript

C++ DEVELOPMENT

STL · Smart pointers · Move semantics · RAII · Algorithms and Data Structures · Design Patterns (GoF) · CMake · OOP · SOLID · DRY

INFRASTRUCTURE

Git · Linux · Docker · PostgreSQL · CI/CD · Kubernetes

NETWORKS / API

REST · HTTP / HTTPS · TCP/IP · DNS / SSH · SSL / TLS · gRPC

LANGUAGE PROFICIENCY

Russian – native

English – B1

confident · basic

Kuropatin Vladislav Sergeevich

C++ Developer

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Work Experience

Sorting Algorithms & Design Patterns — Academic experience

Pet project: GoF design patterns and classical sorting algorithms in C++

STACK: C++17/20 · STL · templates · std::unique_ptr · std::shared_ptr · RAII · SOLID · OOP · GoF Patterns · Algorithms and Data Structures · Visual Studio 2026 · MSVC · Git

Implemented 22 design patterns (GoF) and classical sorting algorithms (QuickSort, MergeSort, BubbleSort, and others) in C++17/20 using template functions, smart pointers (std::unique_ptr, std::shared_ptr), and SOLID principles. Each pattern is placed in a separate project, which provides 22 isolated examples for comparative analysis of architectural solutions. The sorting algorithms are collected in a single module with a comparative analysis of time complexity ($O(n \log n)$ versus $O(n^2)$), which also allows the repository to be used as a reference when choosing a sorting algorithm.

sorting-algorithms-and-design-patterns

AI Insurance Cyber Risk Forecast — Research experience

Forecasting financial losses from cyber incidents

STACK: Python 3.10+ · PyTorch · NumPy · Pandas · SciPy · Scikit-learn · Matplotlib · Seaborn · PyYAML · Pytest · Git

Developed a system for forecasting financial losses from cyber incidents for insurance companies: implemented a complete ML pipeline from generating synthetic data (24 features) and preprocessing to training a multi-head neural network with a shared backbone and three heads: incident probability, conditional loss, and unconditional expected loss. Added a console application with a menu, configurations, unit tests, and visualization of underwriting results.

ai-insurance-cyber-risk-forecast

Mission Titan — Research experience

Numerical modeling of an interplanetary flight

STACK: Python 3.10+ · NumPy · SciPy · Matplotlib · Astropy · poliastro · spiceypy · jplephem · Numba

Developed a numerical N-body gravity interaction model to calculate the flight trajectory of a spacecraft from the Moon to Titan with a gravity assist at Saturn. Implemented an adaptive fourth-order Runge–Kutta method, a solution to Lambert's problem, and initial velocity correction by the shooting method using high-precision NASA SPICE ephemerides. The simulation resulted in a heliocentric velocity reduction of 2.06 km/s and a minimum approach to Titan of 130,034 km.

> A research paper "Numerical Modeling of a Flight to Titan with a Gravity Assist at Saturn" was published in the proceedings of the IV International Interdepartmental Scientific and Technical Conference (Moscow, 2026).

mission-titan · elibrary.ru – publication

Achievements

- Finalist of the international competitive programming championship "RuCode" 2025.
- Semifinalist of the all-Russian MTS True Tech Champ 2025 championship in the competitive programming section.
- Winner (2nd degree diploma) of the hackathon "All-Russian IT Marathon: Digital Tools for Business" 2025, case "Intelligent Sales Manager Assistant".
- 1st degree diploma of the III Annual International Forum of Students and Young Scientists "Security in the Digital Universe", 2026 (developed a numerical integrator for an interplanetary flight trajectory with visualization in Python).
- 2nd degree diploma of the XI Scientific and Technical Conference of Students and Postgraduates of RTU MIREA "Modern Trends in Technological Development", 2026 (developed an AI model in Python to predict the effectiveness of anti-sanction measures for Russian technology companies).